



INFLAMMATORY CONDITIONS OF THE JAWS

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INFLAMMATION IN THE JAWS – OVERVIEW AND MECHANISM

- Most common pathologic condition in the jaws
- Teeth create direct pathways for pathogens and irritants to enter bone
- Body responds with inflammation to eliminate pathogens and repair tissue
- Bone metabolism balance:
 - Osteoblastic bone formation
 - Osteoclastic bone resorption
- Influenced by:
 - Microbial pathogenicity
 - Host immune response
 - Vascularity and time
- Inflammatory conditions form a spectrum with varied clinical and radiologic features

GENERAL CLINICAL FEATURES

- **Cardinal Signs of Inflammation:**

- Redness (*rubor*)
- Heat (*calor*)
- Swelling (*tumor*)
- Pain (*dolor*)
- Loss of function (*functio laesa*)

- **Acute Lesions:**

- Rapid onset
- Intense pain
- Fever
- Noticeable swelling

- **Chronic Lesions:**

- Slow progression
- Mild or dull pain
- Low-grade fever
- Gradual swelling

GENERAL IMAGING FEATURES

- **Location:**

- *Pulpal origin*: Lesion epicenter at tooth apex or along root surface
- *Periodontal origin*: Epicenter at alveolar crest; severe cases may involve furcation
- *Osteomyelitis*: Diffuse, uncontained inflammation; commonly in posterior mandible

- **Periphery:**

- Ill-defined borders
- Gradual transition to sclerotic or radiolucent patterns

GENERAL IMAGING FEATURES-CONTND

- **Radiographic Appearance:**
 - Radiolucent
 - Radiopaque
 - Mixed (combination of both)
- **Disease Progression:**
 - *Acute inflammation:* Bone resorption dominates → radiolucency
 - *Chronic inflammation:* Bone formation increases → radiopacity
- **Osteomyelitis:**
 - Presence of **sequestra**
 - Ill-defined radiolucent areas with radiopaque islands of nonvital bone



GENERAL IMAGING FEATURES-CONTND

Effects on surrounding structures:

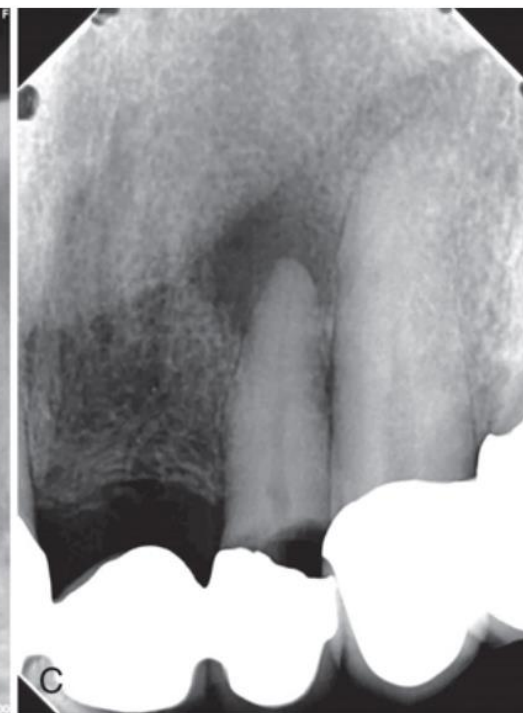
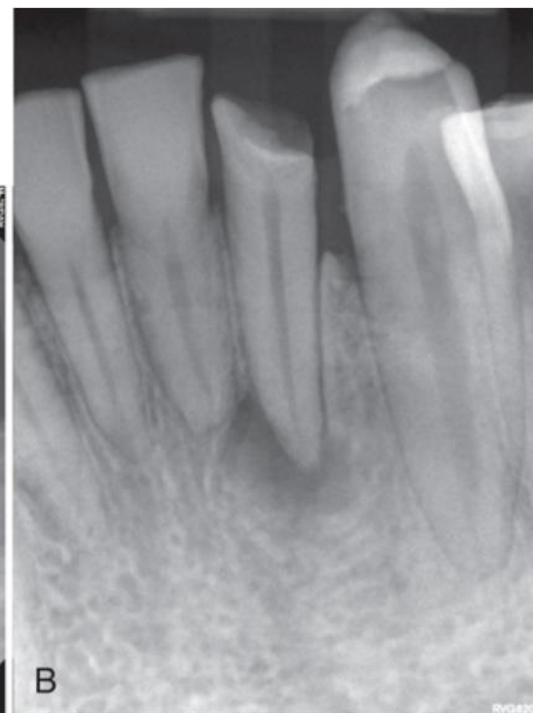
- New bone formation → Sclerotic bone
- Bone resorption → Radiolucent areas
- Periodontal ligament (PDL) widening
 - Common at apex and alveolar crest
- Root resorption
- Periosteal bone formation

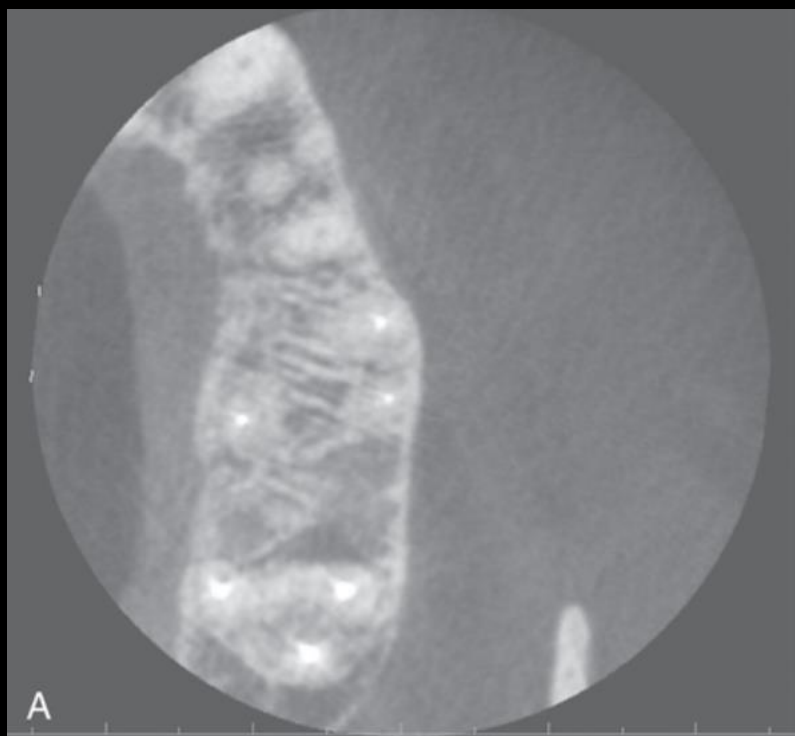


PERIAPICAL INFLAMMATORY LESIONS

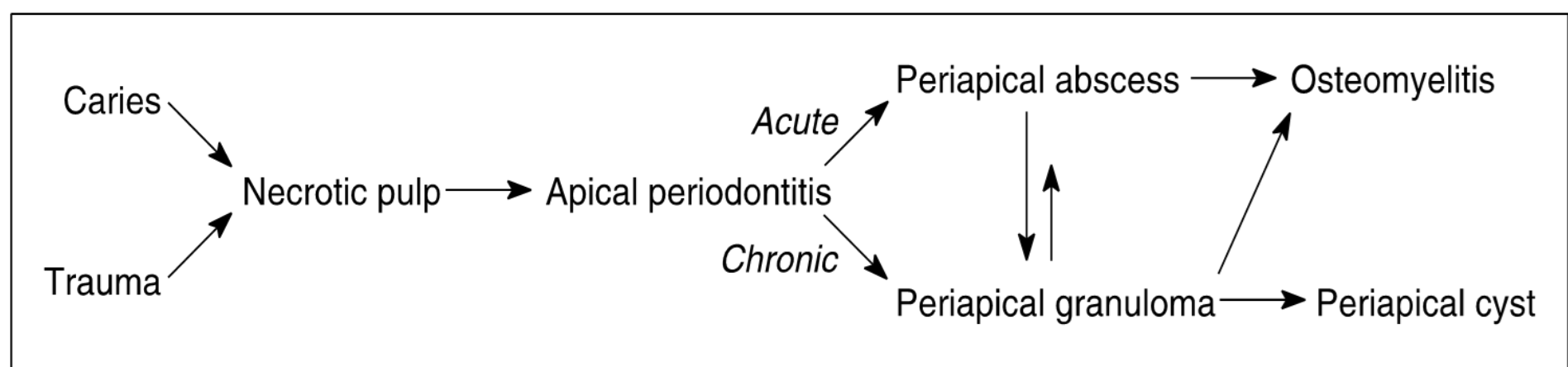
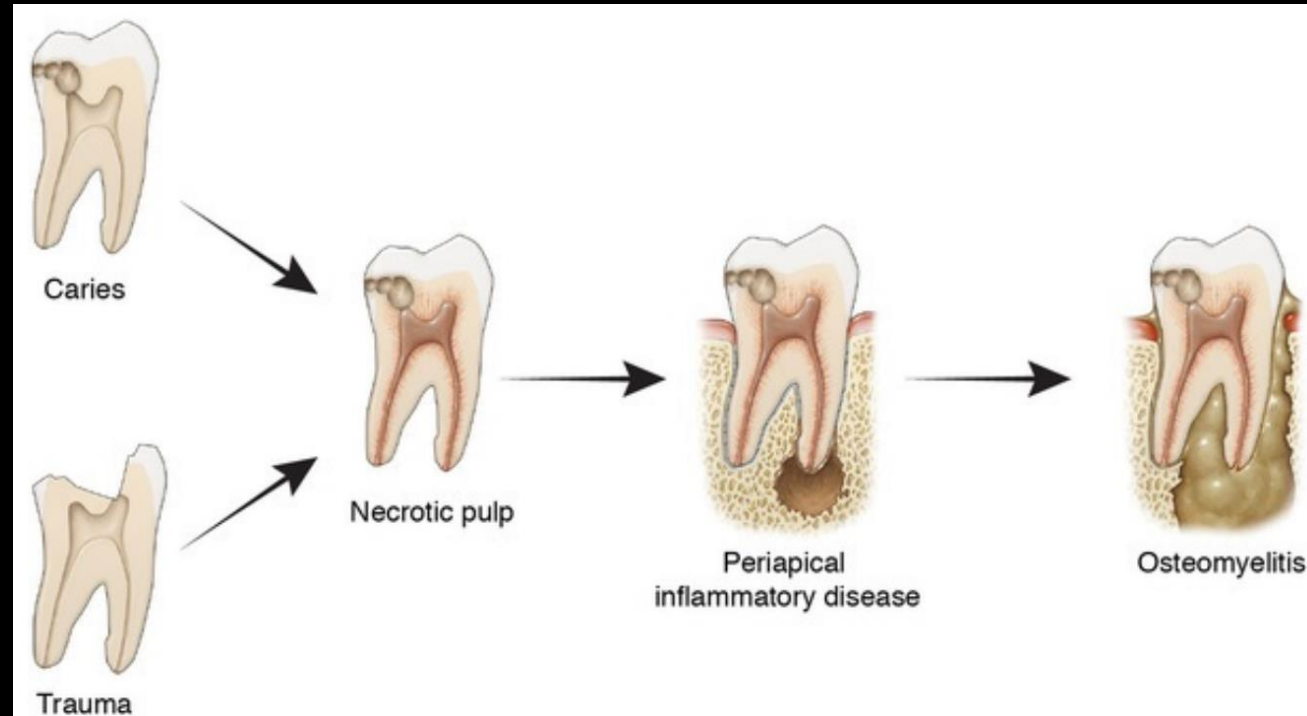
Terminology & Radiologic Diagnosis

- Common Terms: Apical periodontitis, abscess, granuloma, cyst
- Histopathologic Diagnosis: Only microscopic exam can differentiate abscess, granuloma, cyst
- Radiologic Terms:
 - *Rarefying osteitis*: Bone resorption → radiolucency
 - *Sclerosing osteitis*: Bone deposition → radiopacity
- Preferred Terminology: Use “sclerosis” over “condensing/focal sclerosing osteitis”



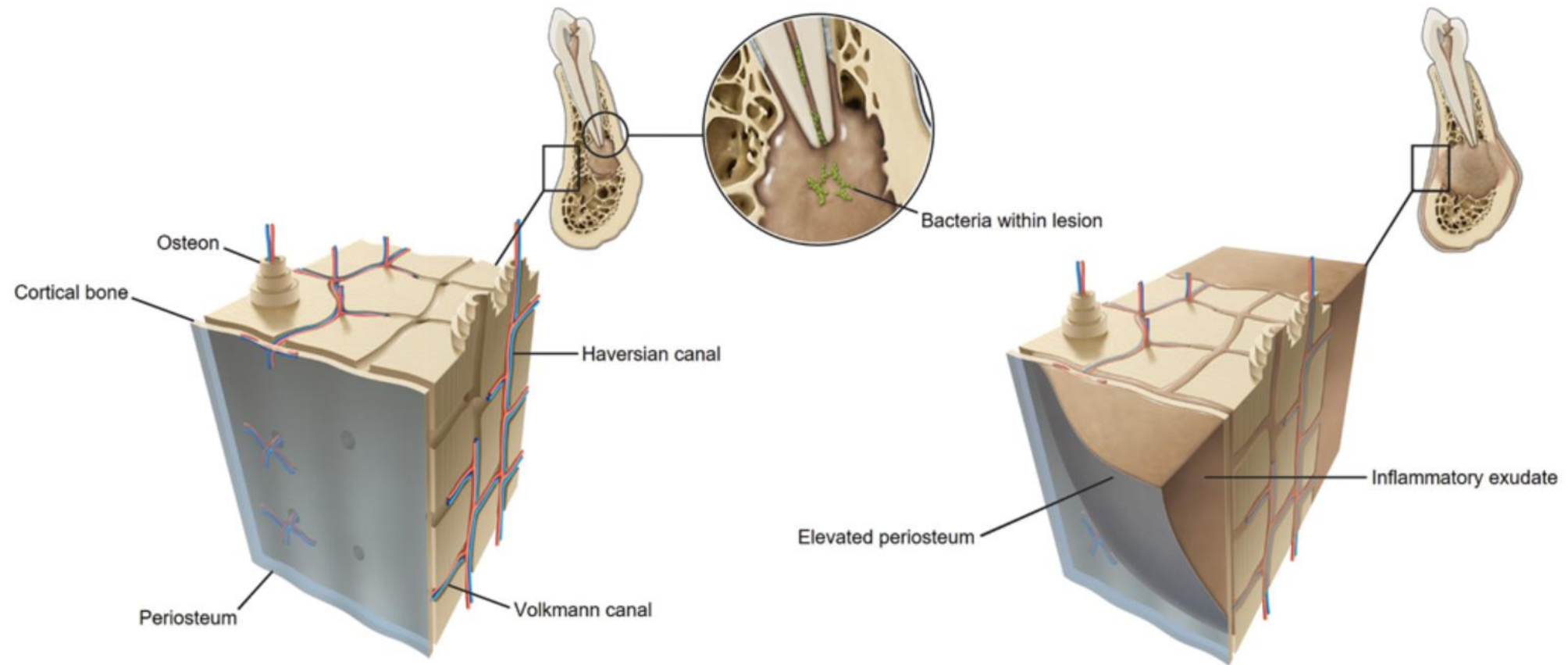


PERIAPICAL INFLAMMATORY LESIONS- DISEASE MECHANISM



PERIAPICAL INFLAMMATORY LESIONS- DISEASE MECHANISM & PATHOPHYSIOLOGY

- **Cause:** Pulpal necrosis from caries, trauma, or chemicals
- **Inflammatory Spread:** From apex to surrounding bone and PDL
- **Progression:**
 - Acute: Abscess formation
 - Chronic: Granuloma or cyst formation
- **Bone Response:**
 - Sclerosis at lesion periphery
 - Periosteal new bone formation
 - Sequestrum formation due to compromised blood supply



Development and progression of periapical inflammatory disease located at the tooth root apex. The inflammatory response spreads through the bone via the Haversian and Volkmann canal systems. As the response reaches the bone surface through Volkmann canals, the exudate lifts and distends the surface periosteum.

PERIAPICAL INFLAMMATORY LESIONS- CLINICAL & IMAGING EXAMINATIONS

- **Clinical Presentation:**
 - Acute: Pain, swelling, mobility, pus drainage
 - Chronic: Asymptomatic or intermittent pain
- **Imaging Modalities:**
 - Periapical, occlusal, panoramic
 - CBCT, MDCT for bone changes
 - MRI for marrow involvement
 - Nuclear medicine (^{99m}Tc , ^{67}Ga) for confirmation

PERIAPICAL INFLAMMATORY LESIONS- IMAGING FEATURES

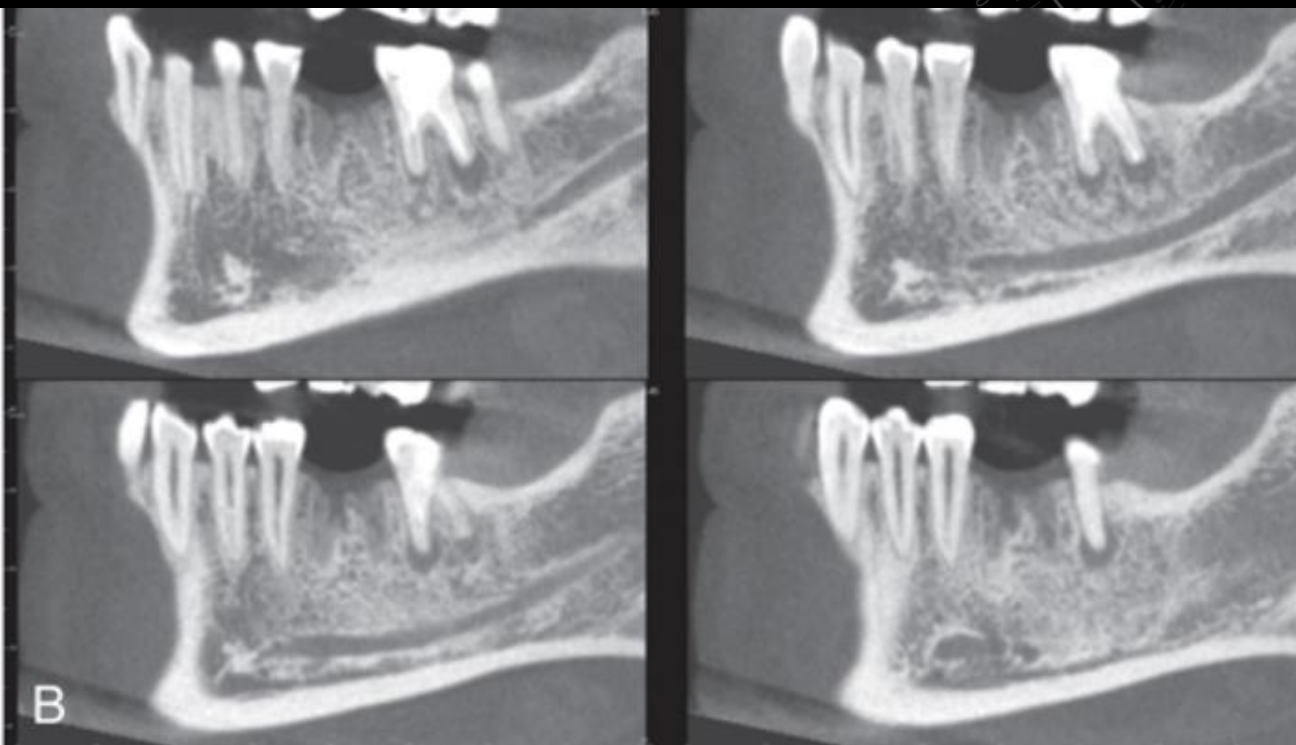
Imaging Features – Location & Periphery

- **Location:**
 - Typically at apex or apical third of tooth root
 - May migrate apically as lesion enlarges
 - Can occur at accessory canals, perforations, or root fractures
- **Periphery:**
 - Poorly to moderately defined borders
 - Smooth contour
 - Transition zone reflects degree of sclerosis
 - Rarely corticated or sharply defined



PERIAPICAL INFLAMMATORY LESIONS- IMAGING FEATURES

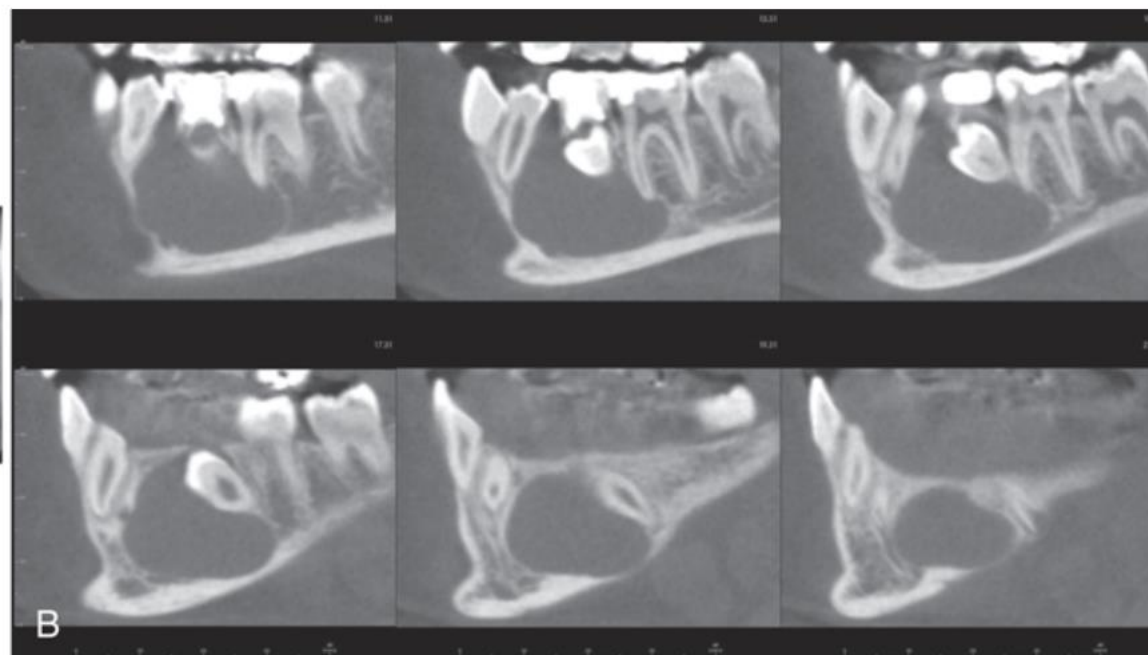
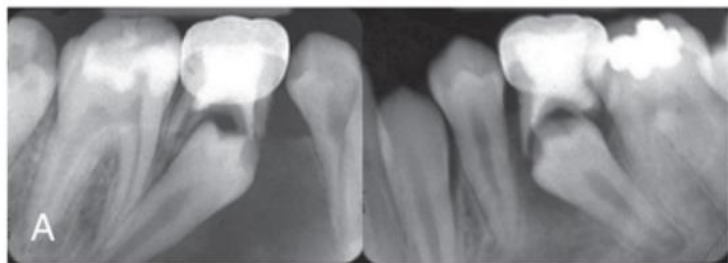
- **Internal Structure:**
 - Early lesions may show no visible changes
 - Initial signs: PDL widening, loss of lamina dura
 - Advanced: Radiolucency at apex, possible sclerosis
- **Effects on Bone:**
 - Bone perforation through cloaca
 - Altered trabecular pattern and marrow space
 - Dense bone with reduced vascularity → sequestrum
 - Periosteal new bone formation: “halo” or “onion skin” patterns



PERIAPICAL INFLAMMATORY LESIONS- IMAGING FEATURES

Effects on Adjacent Teeth & Air Spaces

- **Tooth Effects:**
 - External root resorption → irregular root contour
 - Hypercementosis → bulbous root shape
 - Asymmetric and nonuniform changes
- **Air Space Involvement:**
 - Inflammation may reach sinus or nasal floor
 - Stimulates mucosal thickening → mucositis
- **Deciduous Teeth:**
 - May disrupt eruption of permanent teeth
 - Can cause displacement of developing dentition



PERIAPICAL INFLAMMATORY LESIONS

Differential Interpretation – PCOD vs. Inflammatory Disease

- **PCOD (Periapical Cemento-Osseous Dysplasia):**
 - Early stage: radiolucent, mimics inflammation
 - Tooth vitality test: **vital** pulp
 - Mature stage: central radiopacity within radiolucency
 - Root changes: **hypercementosis**, not resorption
- **Periapical Inflammatory Disease:**
 - Tooth vitality test: **nonvital** pulp
 - Radiopacity at lesion **periphery**
 - Root changes: **external resorption**



PERIAPICAL INFLAMMATORY LESIONS

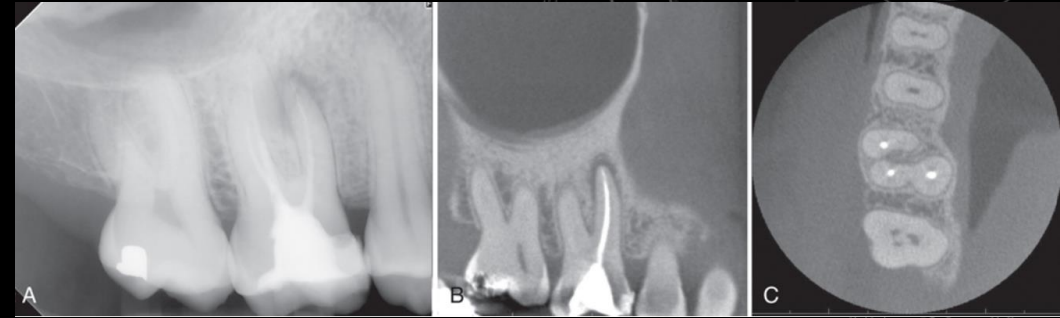
Differential Interpretation – DBI & Malignancy

- **Dense Bone Island (DBI):**
 - May mimic sclerosing osteitis
 - No PDL space alteration
 - Tooth remains **vital**
 - Narrow transition zone to normal bone
- **Malignancy (e.g., leukemia):**
 - Rare, but may involve apical PDL space
 - Multiple areas of cancellous bone destruction
 - Requires further imaging and clinical correlation



PERSISTENT PERIAPICAL INFLAMMATORY DISEASE

- **Causes of Persistence:**
 - Inadequate endodontic treatment
 - Untreated canal, root perforation, or fracture
- **Imaging Clues:**
 - Recurring radiolucency with hydraulic periphery
 - CBCT useful for identifying root fractures or missed canals
- **Management:**
 - Retreatment or periapical surgery
 - Extraction if root fracture is confirmed





OSTEOMYELITIS – OVERVIEW & DISEASE MECHANISM

- **Definition:** Inflammation involving bone marrow
- **Progression:** More severe form of periapical inflammatory disease
- **Risk Factors:** Immunosuppression, systemic disease, poor vascularity
- **Mechanism:**
 - Spread from nonvital tooth or hematogenous origin
 - Acute: neutrophilic infiltrate
 - Chronic: fibrosis, mononuclear cells

CLINICAL & IMAGING FEATURES OF OSTEOMYELITIS

- **Clinical Features:**
 - Acute: pain, swelling, fever, lymphadenopathy, paresthesia
 - Chronic: intermittent symptoms, fistula, possible asymptomatic
- **Imaging Features:**
 - Sclerosis, cloaca, periosteal new bone formation
 - Sequestra: necrotic bone
 - “Onion skin” or “proliferative periostitis” in children
- **Imaging Modalities:**
 - Periapical, panoramic, occlusal
 - CBCT/MDCT for bone changes
 - MRI for marrow; ^{99m}Tc & ^{67}Ga for inflammation

OSTEOMYELITIS -LOCATION, PERIPHERY & PROGRESSION

- **Location:**
 - Most common in posterior mandible
 - Rare in maxilla
- **Periphery:**
 - Acute: poorly defined, noncorticated
 - Chronic: better defined with sclerotic transition zone
- **Progression:**
 - Cortical bone resorption
 - Periosteal stimulation → new bone layers
 - Multiple “onion skin” layers in children

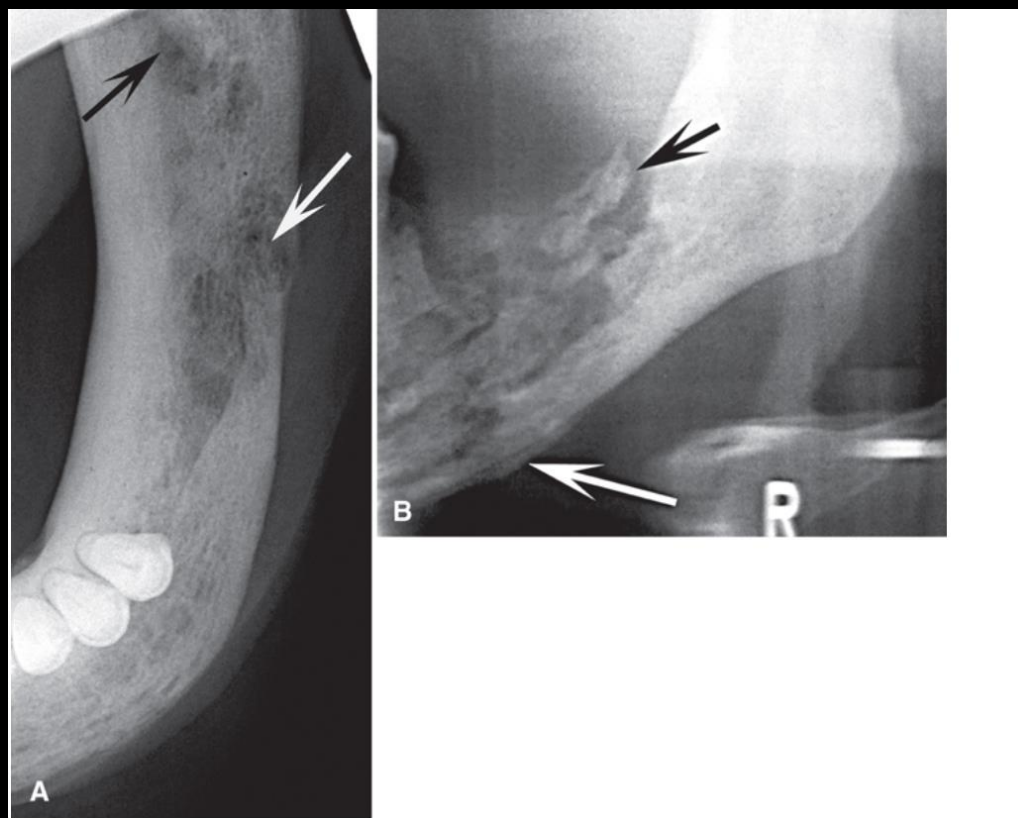


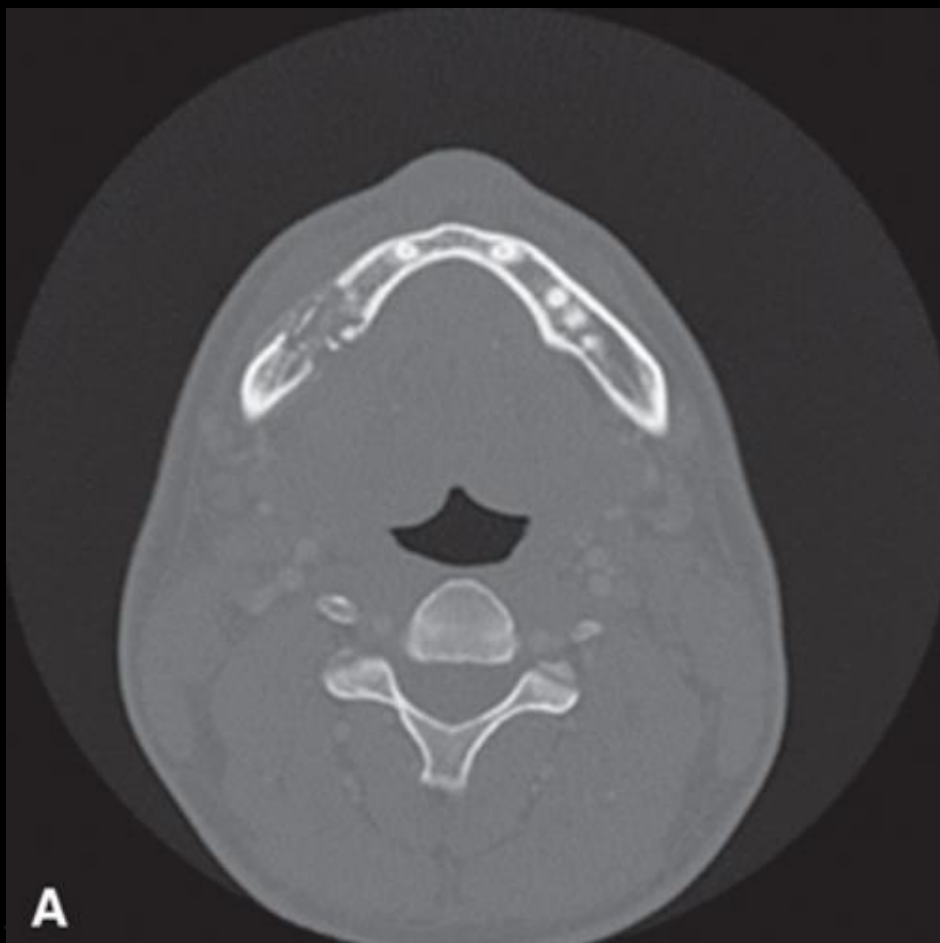


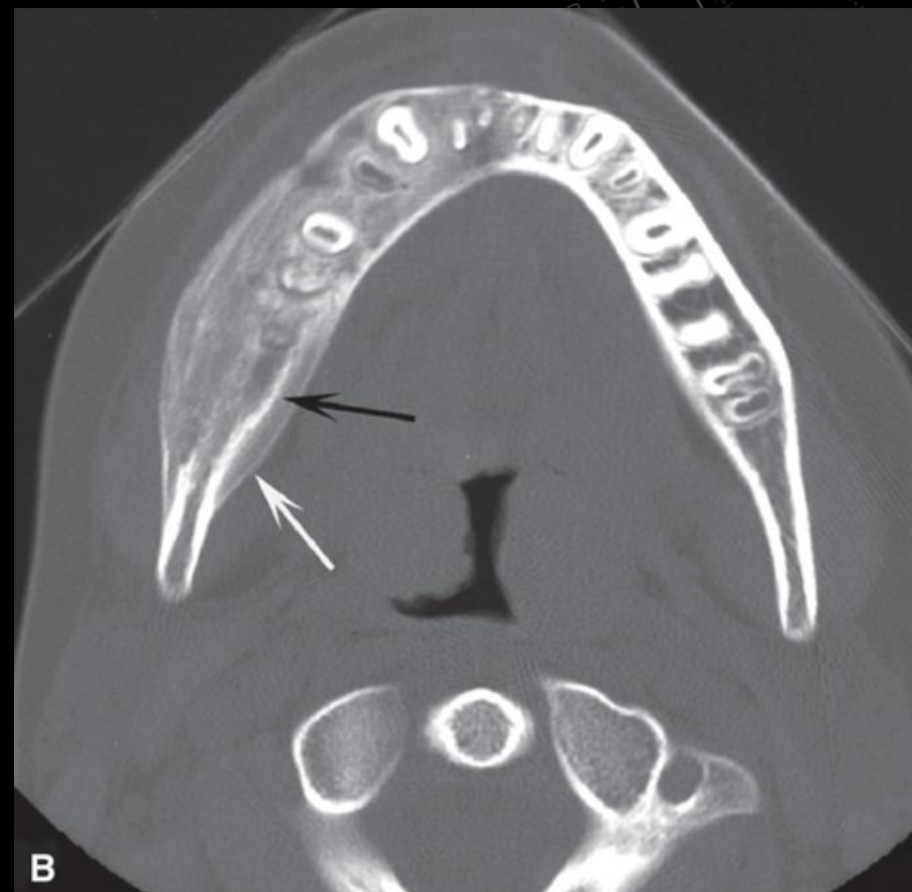
INTERNAL STRUCTURE OF OSTEOMYELITIS

- **Acute Phase:**
 - Decreased bone density
 - Loss of trabecular definition
 - Focal radiolucency in scattered regions
- **Chronic Phase:**
 - Increased bone density → uniform radiopacity
 - Appearance of **sequestra**:
 - Radiopaque bone fragments
 - Surrounded by radiolucent halo
- **Imaging Tools:**
 - CBCT/MDCT superior for detecting sequestra and dense bone changes









OSTEOMYELITIS - EFFECTS ON SURROUNDING STRUCTURES

- **Periosteal Reaction:**
 - New bone forms in elevated periosteum
 - Granular/sclerotic bone pattern develops
- **Advanced Changes:**
 - Loss of original cortex visibility
 - Bony asymmetry and surface contour changes
 - Formation of **cloaca** (cortical break)
- **Spread:**
 - May involve TMJ → septic arthritis
 - Possible extension to mastoid air cells and inner ear

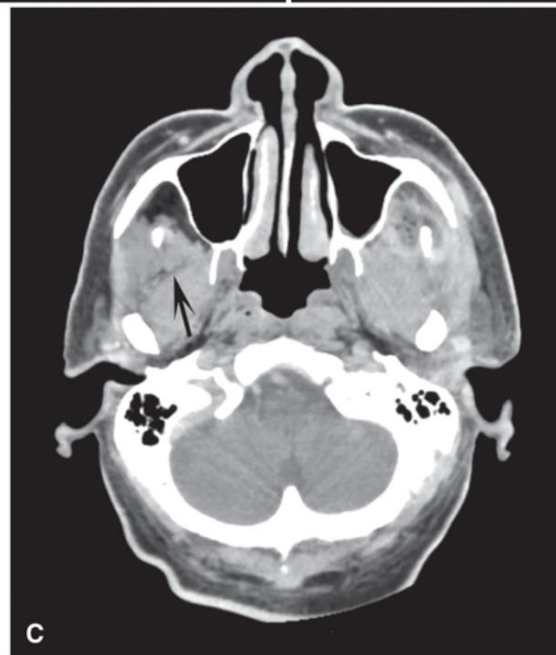
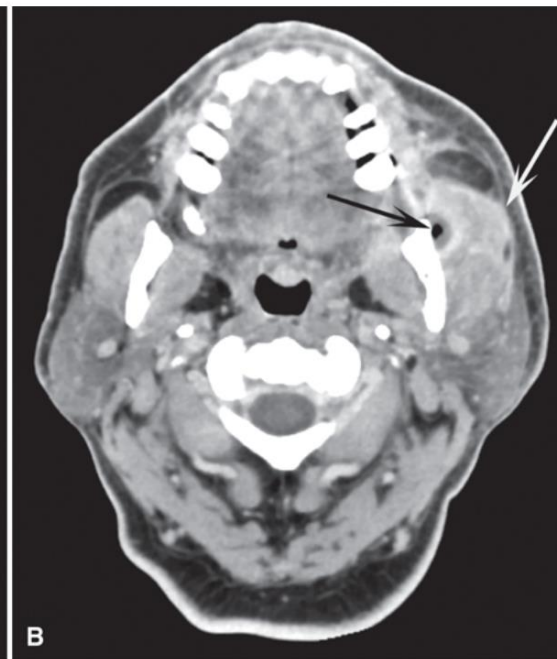
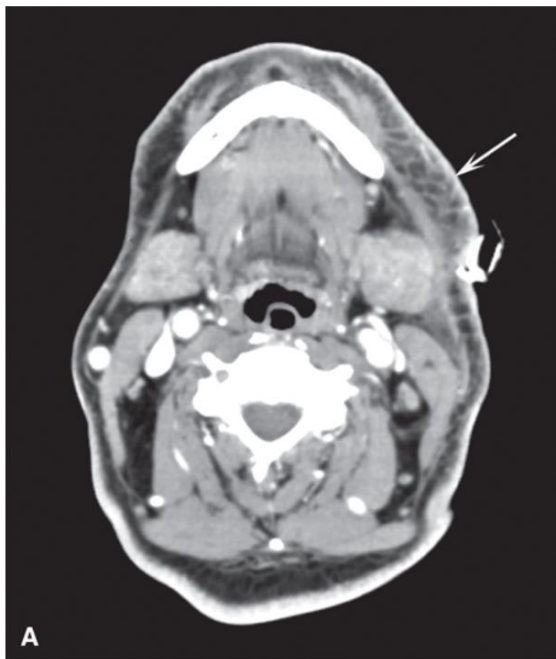
OSTEOMYELITIS - EFFECTS ON TEETH & SOFT TISSUE IMAGING

- **Tooth Effects:**

- External root resorption
- Hypercementosis
- Widened PDL space against sclerotic bone

- **Soft Tissue Imaging:**

- MDCT with contrast: “fat streaking”
- MRI with gadolinium: edema, abscess, lymphadenopathy
- Abscess: low attenuation center with enhanced border



OSTEOMYELITIS - DIFFERENTIAL DIAGNOSIS

- **Osteomyelitis:**
 - Periosteal new bone formation
 - Sequestra present
 - Cortex may be indistinct in chronic phase
- **Fibrous Dysplasia:**
 - Bone enlarges from within
 - Cortex remains intact but thinned
 - No sequestra formation
- **Key Differentiator:** Location and origin of new bone formation

OSTEOMYELITIS - DIFFERENTIAL DIAGNOSIS

Osteomyelitis vs. Malignancy

- **Malignancy:**
 - May mimic acute osteomyelitis
 - Often secondarily infected
 - Destruction of adjacent structures
 - Soft tissue mass and rapid growth
- **Radiologic Clues:**
 - Langerhans cell histiocytosis: radiolucent, no sclerosis
 - Ewing sarcoma: “onion skin” periosteal reaction
 - Osteosarcoma/chondrosarcoma: intralesional bone formation
- **Paget Disease:** Bilateral mandibular involvement, no sequestra

OSTEOMYELITIS - DIFFERENTIAL DIAGNOSIS

Chronic Variants & Syndromic Mimics

- **Chronic Osteomyelitis:**
 - May evolve from acute phase
 - Negative cultures in chronic stage
- **Primary Chronic Osteomyelitis:**
 - No identifiable pathogen
 - Similar to CRMO and SAPHO syndrome
- **SAPHO Syndrome** (Synovitis, acne, pustulosis, hyperostosis, osteitis)
 - Imaging: sclerosis, swelling, negative cultures
- **CRMO (Chronic Recurrent Multifocal Osteomyelitis):**
 - Symmetric long bone involvement in children
 - Nonpurulent, culture-negative
 - Jaw involvement may be unifocal variant



OSTEOMYELITIS - MANAGEMENT

- Acute: remove source, antibiotics, surgical drainage
- Chronic: may require sequestrectomy, decortication, resection
- Limited success with antibiotics in sclerotic bone
- Alternative therapies: steroids, NSAIDs, bisphosphonates
- Hyperbaric oxygen therapy in select cases

RADIATION-INDUCED BONE CHANGES – MECHANISM & EFFECTS

- **Cause:** Therapeutic radiation for head and neck malignancies
- **Cellular Damage:** DNA, proteins, lipids → impaired cell function or death
- **Dose-Dependent Effects:**
 - Low dose (1–4 Gy): osteoblast stimulation → sclerosis
 - Moderate dose (10–20 Gy): cell death, growth retardation
 - High dose (60–70 Gy): mature bone damage
- **Vascular Injury:** Hypovascular, hypoxic bone → impaired healing
- **Outcome:** Osteoradionecrosis (bone death due to radiation)

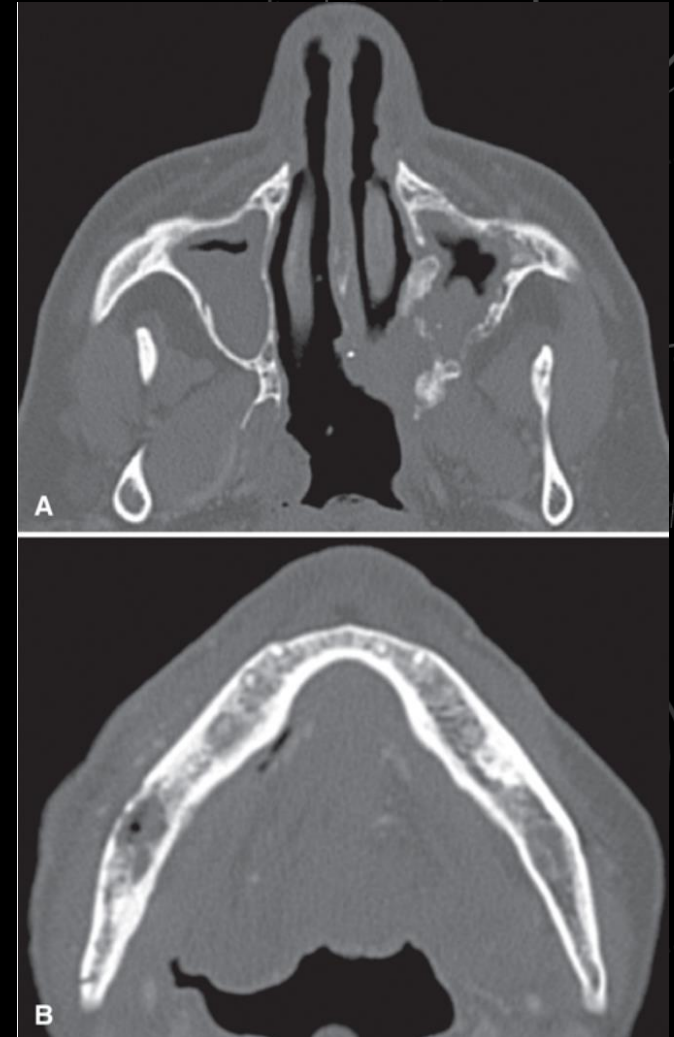
IMAGING FEATURES & DIFFERENTIAL INTERPRETATION

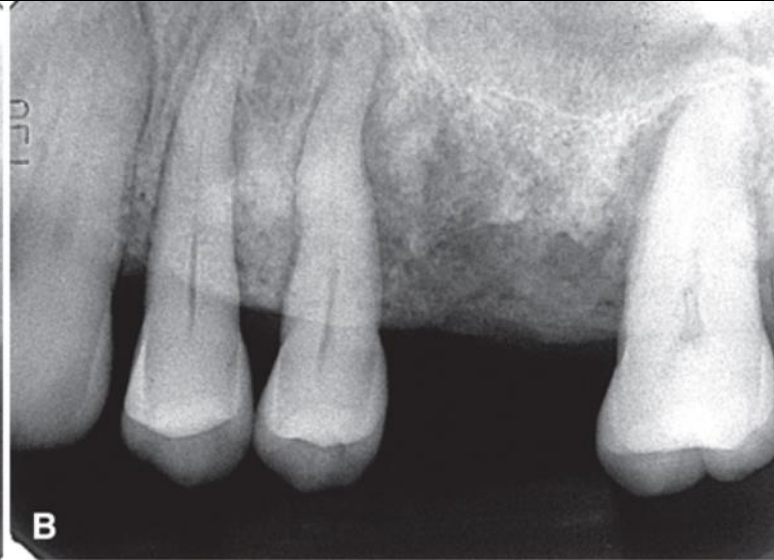
- **Imaging Features:**

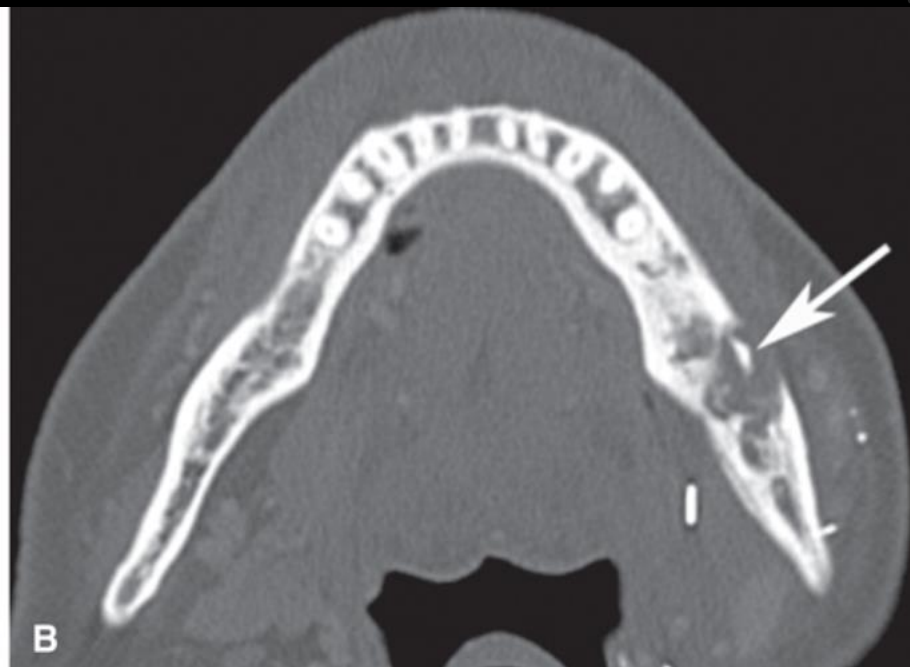
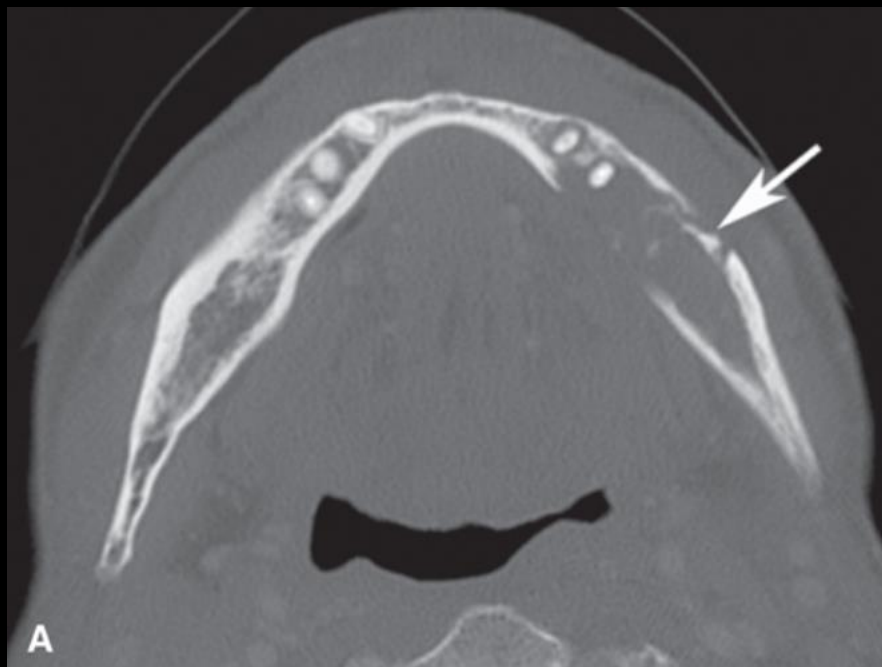
- Early: no visible changes
- Later: cortical bone resorption, sclerosis, sequestra
- PDL space widening (uniform, no epicenter)
- Possible jaw fractures

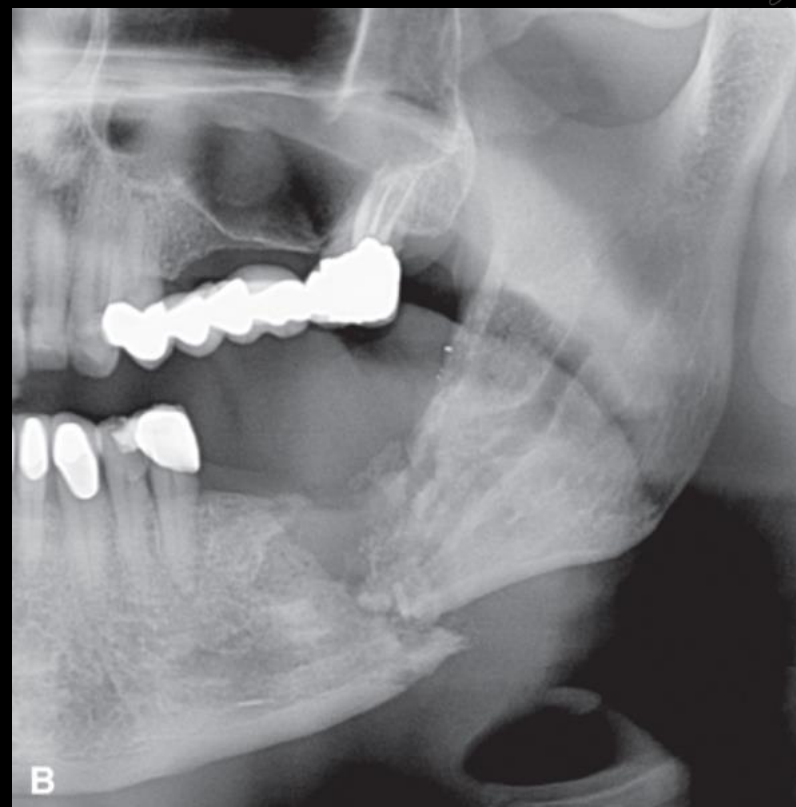
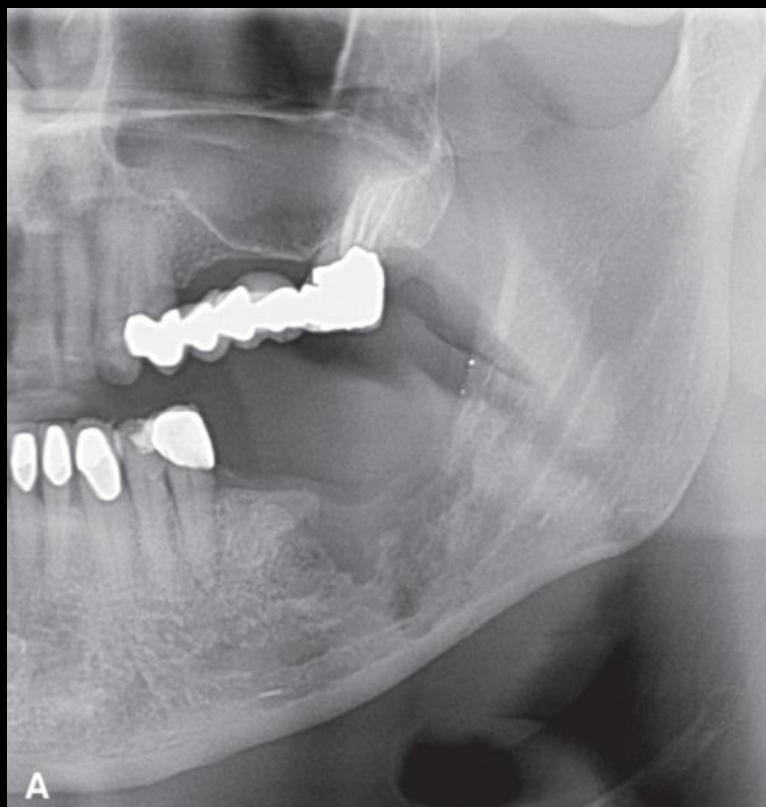
- **Differential Diagnosis:**

- Differentiate from periapical disease (epicenter at apex)
- Periodontal disease: changes at alveolar crest
- Malignancy: soft tissue mass, irregular bone destruction
- Use CBCT/MDCT for detailed evaluation









CLINICAL FEATURES & MANAGEMENT

- **Clinical Signs:**
 - Exposed bone, pain, swelling, drainage
 - May be asymptomatic
 - Risk of fracture and secondary infection
- **Histopathology:**
 - Absence of osteocytes in lacunae
 - Loss of mucosal covering
- **Management:**
 - Prevention: avoid trauma post-radiation
 - Imaging: panoramic, CBCT/MDCT
 - Treatment: conservative care, possible surgical intervention
 - Monitor for malignancy recurrence with MRI/MDCT

MRONJ – DISEASE MECHANISM & RISK FACTORS

- **Cause:** Chemical trauma from medications
- **Key Drugs:**
 - Bisphosphonates
 - RANK-L inhibitors
- **Mechanism:**
 - Inhibits osteoclasts → reduced bone resorption
 - Leads to necrotic bone exposure, especially post-surgery
- **High-Risk Scenarios:**
 - IV administration
 - Potent aminobisphosphonates
 - Dental extractions, implants, periodontal surgery

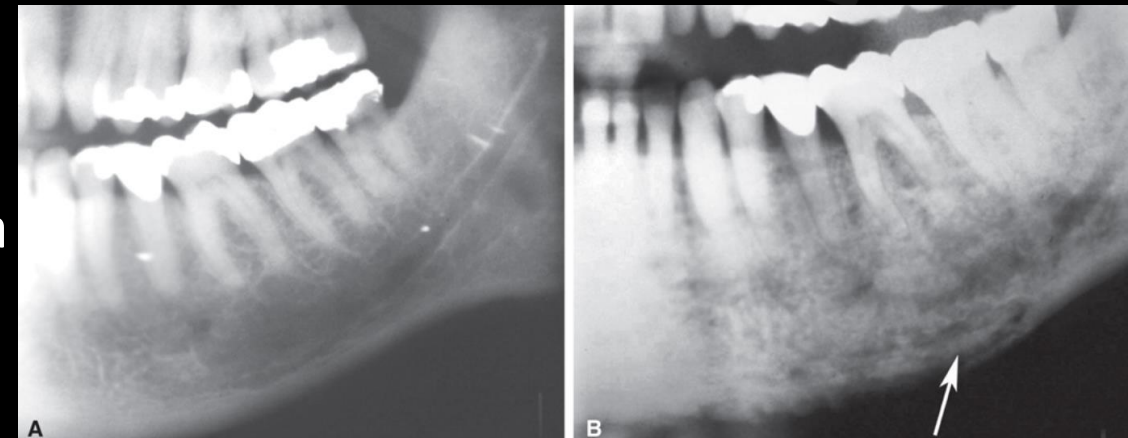
CLINICAL & IMAGING FEATURES OF MRONJ

- **Clinical Presentation:**

- Exposed bone post-surgery or trauma
- Common sites: posterior mandible (60%), maxilla (40%)
- May be asymptomatic or painful with swelling

- **Imaging Features:**

- May resemble osteomyelitis
- Sequestra formation
- Widened PDL space, thickened lamina dura
- Rapid and dramatic changes possible



MANAGEMENT & PREVENTION OF MRONJ

- **Prevention is Key:**
 - Dental evaluation before starting IV therapy
 - Remove infection sources to avoid future surgery
- **Challenges in Treatment:**
 - Surgical and hyperbaric oxygen therapy often ineffective
 - Long drug half-life (up to 12 years) complicates recovery
- **Symptom Management:**
 - Antibiotic mouth rinses
 - Systemic antibiotics
 - Pain control
- **Emerging Therapies:**
 - Steroids, NSAIDs
 - Bisphosphonate therapy adjustments



PERICORONITIS – DISEASE MECHANISM & CLINICAL FEATURES

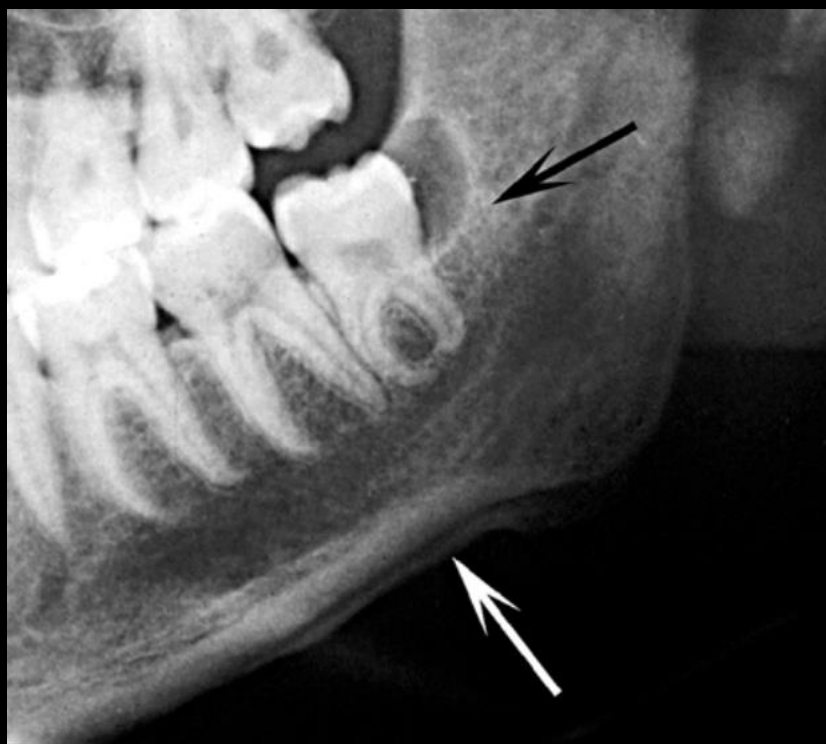
- **Definition:** Inflammation of soft tissue operculum over a partially erupted tooth
- **Common Site:** Mandibular third molars in young adults
- **Cause:** Food/microbial debris trapped under operculum
- **Symptoms:**
 - Pain, swelling, trismus
 - Ulcerated operculum, pain on occlusion

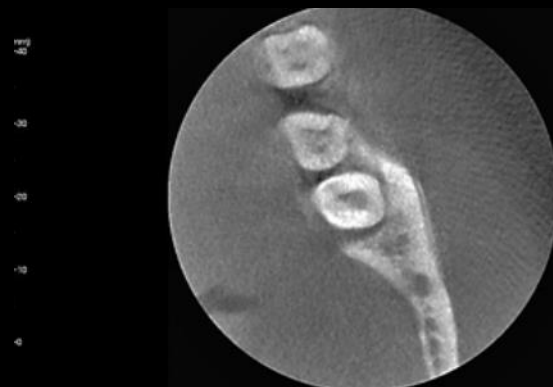
IMAGING FEATURES & DIFFERENTIAL INTERPRETATION

- **Imaging Findings:**
 - May show no changes if confined to soft tissue
 - Bone rarefaction or sclerosis near crown/root
 - Severe cases may resemble osteomyelitis
- **Location:** Follicular space near embedded crown
- **Periphery:** Poorly defined with wide transition zone
- **Differential Diagnosis:**
 - Dense Bone Island (DBI), fibrous dysplasia
 - Squamous cell carcinoma, sclerotic osteosarcoma
 - Clinical symptoms help exclude non-inflammatory conditions

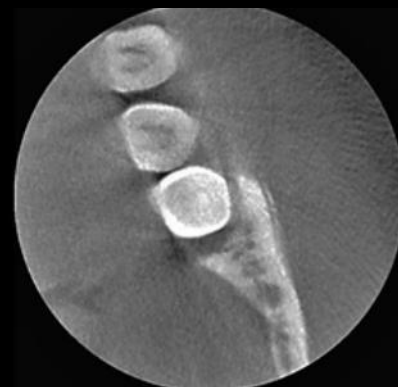
MANAGEMENT & EFFECTS ON SURROUNDING STRUCTURES

- **Effects on Bone:**
 - Rarefaction and sclerosis
 - Possible periosteal new bone formation
 - May extend to sigmoid notch in severe cases
- **Treatment:**
 - Goal: remove partially erupted tooth
 - Acute phase: antibiotics to manage symptoms
 - Definitive treatment delayed if trismus is present

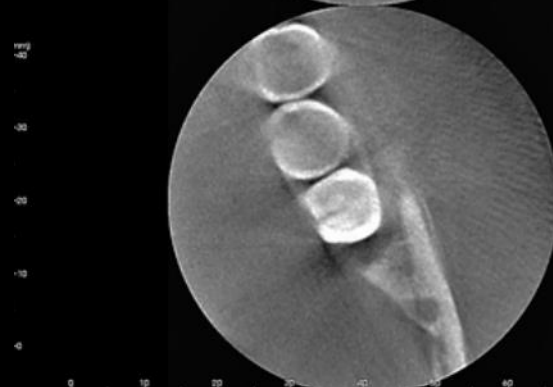




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